

Exploring Alpha Generation through Fixed Income Arbitrages

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Motivation and Research Questions

Research Motivation

- Fixed income arbitrages remain underexplored compared to equity "factor zoo"
- Persistent mispricings challenge market efficiency assumptions
- Key puzzle: Why do textbook arbitrages persist despite sophisticated investors?

Core Issue

Are observed returns compensation for systematic risks (risk premia) or true market inefficiencies (alpha)?

Novelty

First unified analysis of 4 arbitrages with novel datas and modern econometrics and ML techniques

Research Questions

- ① **RQ1:** Which risk factors systematically explain excess returns of fixed income arbitrage strategies?
- ② **RQ2:** Do these strategies generate statistically & economically significant alpha after risk adjustment?
- ③ **RQ3:** Are there interdependencies suggesting slow-moving arbitrage capital (Duffie 2010)?

Key Differentiator

Combines friction-based arbitrage theory with modern asset pricing methods

The Four Arbitrage Strategies

Strategy 1: CDS-Bond Basis Arbitrage

Basis Formula

$$\text{CDS Basis}_t = \text{CDS Spread}_t - \text{ASW Spread}_t$$

- Long (short) bond in ASW and buy (sell) CDS protection on the same issuer and maturity.
- **If no default:** earn the spread differential (basis) between bond and CDS.
- **If default:**
 - Bond pays recovery value R , CDS pays $100 - R \rightarrow \text{total} = \text{par}$
- *Perfect hedge: arbitrage profit realized regardless of default outcome.*

Strategy 2: Inflation-Linked Bond Arbitrage

Basis Formula

$$\text{Basis}_t = y_t^{\text{IL BTP}} - y_t^{\text{Nominal BTP}}$$

- Long (short) Italian inflation-linked bond (BTP€i or BTP Italia).
- Enter inflation swap to transform inflation-linked cash flows into fixed.
- Short (long) nominal BTP with matched maturity.
- **If held to maturity:** realize the basis implied by the relative mispricing of the inflation-linked bond.

Data Innovation

First study of italian inflation bonds vs BTP arbitrage

Strategy 3: CDS Index Skew Arbitrage

Skew Formula

$$\text{Skew}_t = \text{Market Spread}_t^{\text{CDSIndex}} - \text{Fair Value Spread}_t^{\text{CDSIndex}}$$

The fair value spread is the duration-weighted average of the single-name CDS fair spreads.

- Long (short) protection on index, short (long) protection on weighted single-names (or vice versa).
- **If no default:** net coupon flow equal to spread difference (skew).
- **If default:** protection payments match across index and constituents.
- *Cash flows are perfectly matched: hedge is delta-neutral and riskless.*

Theoretical Basis

Violation of linearity property: $\text{Index} \neq \sum \text{Constituents}$

Strategy 4: Tranche vs Single-Name Arbitrage

Value Decomposition Violation

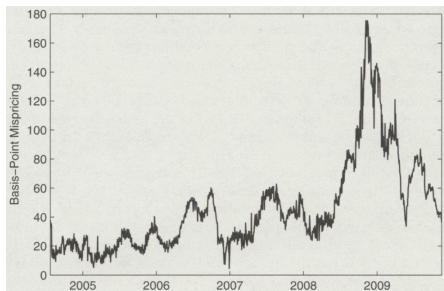
$$\text{Price}_t^{0-100\% \text{ Tranche}} \neq \text{Price}_t^{\text{Single Names CDS portfolio}}$$

- Long (short) the 0–100% tranche on a CDS index.
- Short (long) protection on underlying single-name CDS in same proportion.
- **If no default:** net coupon stream reflects spread discrepancy.
- **If default:** losses on names are absorbed by tranche, matched by CDS payouts.
- *Perfect replication of losses ensures arbitrage with bounded risk.*

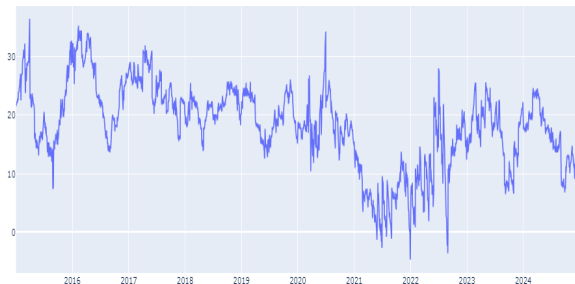
Literature Review and Contributions

Fleckenstein, Longstaff & Lustig (2014)

- Using data from 2004–2009 of TIPS vs Treasuries arbitrage, they find:
 - **Average mispricing: 60 bps**, peaking above **200 bps** during the financial crisis.
 - Treasuries were persistently overpriced relative to synthetic versions (TIPS + inflation swaps).
- Reject standard explanations: neither liquidity, credit risk, repo frictions, nor transaction costs can explain the mispricing.
- **Today, this arbitrage has disappeared**: basis is compressed to **a few basis points only**.



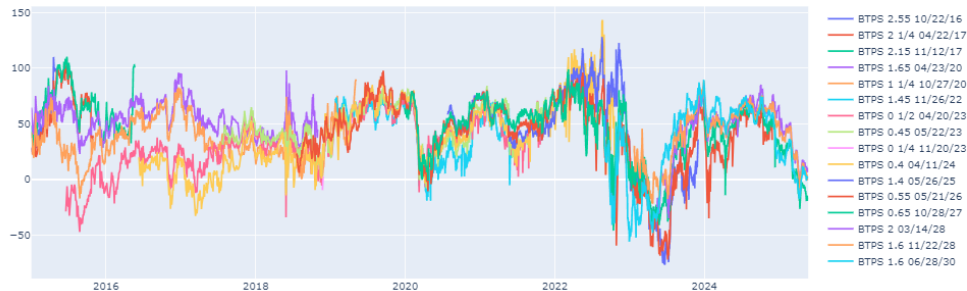
TIPS Basis (2004–2009), Fleckenstein et al.



TIPS Basis (2015–2025), own calculation

Fleckenstein, Longstaff & Lustig (2014)

- Unlike the TIPS market, arbitrage opportunities **still persist** in Italian ILBs, despite their high liquidity.
- This arbitrage **cannot be explained** by:
 - Repo rates (same as nominal BTPs).
 - Ballooning effects (absent in BTP Italia due to immediate revaluation).
- Data innovation: First study of Italian inflation bonds vs BTP dynamics



Duarte, Longstaff & Yu (2007)

- Among the first to test whether fixed-income arbitrage strategies earn alpha.
- Analyze six strategies: swap spread, yield curve, mortgage, volatility, and capital structure arbitrages.
- Find that some strategies generate **significant alpha** even after controlling for standard equity and bond market risk factors.

Empirical Specification

$$R_{it} = \alpha + \beta_1 RM_t + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 UMD_t + \beta_5 RS_t \\ + \beta_6 R2_t + \beta_7 R5_t + \beta_8 R10_t + \beta_9 RI_t + \beta_{10} RB_t + \epsilon_t$$

Factors:

- RM, SMB, HML, UMD: equity market, size, value, and momentum (Fama-French-Carhart).
- RS: S&P bank excess returns.
- R2, R5, R10: excess returns on Treasury portfolios (2Y, 5Y, 10Y).
- RI, RB: excess returns on Merrill Lynch industrial and bank bond indices.

Critical Assessment of Duarte, Longstaff & Yu (2007)

- **Strengths:**

- First paper to evaluate performance of fixed-income arbitrage strategies.
- Shows that some strategies generate alpha net of equity and bond market risk.

- **Limitations / Gaps**

- **No true “textbook” arbitrage:** Most strategies analyzed involve convergence risk and are sensitive to market timing.
- **Fixed betas:** Risk exposures are not time-varying.
- **Arbitrary factor choice:** Regression uses predefined risk factors; model may omit relevant pricing sources.

- **Our Contribution:**

- Study four **textbook arbitrage arbitrages** with guaranteed convergence by construction.
- Address time-variation in risk exposures and formalize factor selection.
- Extend beyond U.S. context to highly liquid European markets.

Sadka (2010): Liquidity Risk in Hedge Funds

- Analyzes whether **systematic liquidity risk** helps explain returns of hedge funds engaging in fixed income arbitrage.
- Rather than examining individual strategies, the study focuses on **portfolio-level returns of arbitrage-oriented funds**.

$$R_{it} = \alpha + \beta_1(MKT - RF)_t + \beta_2SMB_t + \beta_3\Delta TERM_t + \beta_4\Delta CREDIT_t + \beta_5PTFSBD_t \\ + \beta_6PTFSFX_t + \beta_7PTFSCOM_t + \beta_8Liquidity_t + \epsilon_t$$

Factors:

- $(MKT - RF)$: equity market excess return; SMB : size premium.
- $\Delta TERM$, $\Delta CREDIT$: changes in term spread and credit spread.
- $PTFSBD$, $PTFSFX$, $PTFSCOM$: trend-following returns in bond, currency, and commodity futures (Fung & Hsieh, 2002).
- $Liquidity$: innovation in market-wide liquidity (Sadka, 2006).

Our Contributions

- Study **four textbook arbitrages** with guaranteed convergence structure.
- Use **two-step factor testing**: first classical models (Duarte, Longstaff & Yu (2007), Sadka (2010) (Fung & Hsieh, 2002), then ML-driven selection (Elastic Net).
- Model comparison with Barillas–Shanken Bayes Factor (2018).
- Estimate **time-varying betas** using Kalman Filter and regime shifts.
- Test alpha using **robust bootstrap** and Adapting Rolling-Window Regressions (Fama & French, 2010), beyond standard OLS.
- Study **interaction across strategies** to reveal capital constraints and substitution effects (Duffie, 2010).

Research Methodology

Stage 1: Penalized Factor Selection

Elastic Net Optimization

$$\min_{\beta \in \mathbb{R}^p} \|\mathbf{R} - \mathbf{X}\beta\|_2^2 + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|_2^2$$

- Inputs: 40+ candidate factors
 - Legacy factors from literature
 - Novel macro/liquidity/credit variables
- Output: Sparse set of 5-6 relevant factors per strategy

Advantage over Literature

Data-driven selection avoids ad hoc factor choices

Model Comparison: Bayesian Factor Evaluation

Barillas–Shanken Bayes Factor (2018)

$$BF_{10} = \frac{P(\text{Data} \mid H_1)}{P(\text{Data} \mid H_0)}$$

- H_0 : Benchmark factor model (Duarte, Longstaff & Yu (2007), Sadka (2010), Fung & Hsieh, 2002)
- H_1 : Benchmark + Elastic Net-selected strategy-specific factors
- Objective: Assess marginal contribution of selected factors

Interpretation

High BF_{10} supports inclusion of additional risk factors beyond standard models.

Stage 2: Time-Varying Factor Model

$$r_{it} = \alpha_t + \sum_{j=1}^k \beta_{j,t} x_{j,t} + \epsilon_t$$

$$\beta_{j,t} = \beta_{j,t-1} + \delta_j \cdot \text{State}_t + \nu_{j,t}, \quad \nu_{j,t} \sim \mathcal{N}(0, Q)$$

State_t captures time-specific market regimes and event-driven shifts (e.g., VIX spikes, crisis dummies).

- Kalman filter estimation
- Observable regime shifts
- Q-matrix controls beta variability

Why not Markov Switching?

Latent regime estimation (à la Hamilton, 1989) is unreliable with short samples ($T \approx 120$). Observable states (e.g., VIX, policy dummies) allow for robust, interpretable regime dynamics.

Stage 3: Robust Alpha Testing

Bootstrap Procedure

- 1 Estimate model residuals $\hat{\epsilon}_t$ (Politis & Romano 1994)
- 2 Resample blocks $\{\epsilon_t^{(b)}\}$ preserving autocorrelation
- 3 Re-estimate $\alpha^{(b)}, \beta_t^{(b)}$ via Kalman filter
- 4 Repeat 10,000 times for empirical distribution

Rolling-Window Regressions (Fama & French, 2010)

- Windows adapt to local return volatility
- Complemented by **rolling-window regressions** (Fama & French, 2010)
- Tracks regime-dependent behavior of α_t
- Robust to nonstationarity and model instability

Stage 4: Strategy Interdependence and Capital Substitution

- Tests whether returns of arbitrage strategies move together.
- Captures potential reallocation of constrained capital across strategies.
- Estimated via pairwise regressions:

Specification

$$r_t^{(s)} = \alpha + \sum_{j \neq s} \gamma_{sj} r_t^{(j)} + \varepsilon_t$$

Each strategy return is regressed on contemporaneous returns of the other three. Significant γ_{sj} indicates substitution or correlation under shared constraints.

Theoretical Link

Consistent with Duffie (2010): slow-moving capital reallocates gradually, creating staggered arbitrage closures.

Expected Results & Literature Contributions

Expected Results

- **Heterogeneous risk factors:** Different drivers per strategy
 - Inflation arbitrage: 5y5y Inflation swap + VIX
 - CDS-bond basis: Liquidity + eligibility + market segmentation
- **Significant alphas:** For ≥ 1 strategies after risk adjustment
- **Time-varying exposures:** Regime-dependent betas
- Evidence for Duffie's (2010) **slow-moving capital hypothesis**

Novel Contributions to Literature and Practice

- First unified study of Four Textbook Fixed Income Arbitrages with guaranteed convergence
- First study of Italian inflation bonds vs BTP dynamics
- Combining literature with machine learning techniques for factor selection
- Time-varying framework with State-space model and **observable** regime shifts
- Bayesian model comparison (Barillas-Shanken 2018)
- Cross-strategy correlation analysis
- New evidence on mispricing persistence
- Optimal capital allocation

Research Timeline

Project Schedule

Period	Activity
Jun-Jul 2025	Data collection & strategy return construction
Aug 2025	Factor selection (Elastic Net)
Sep 2025	Time-varying estimation (Kalman filter)
Oct 2025	Robustness checks & Bayesian tests
Nov 2025	Writing & presentation

Current Status

Preliminary data collection completed for Italian BTP basis

Thank You!

Questions & Discussion

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